Centrifugal Pump P-207
Mechanical Seal Replacement

Equipment manual, section 7: inspection, pre-fit checks, installation and commissioning of the mechanical seal

Equipment tag	P-207 (duty) – standby pump P-207B, same seal arrangement	**Revision**	C – 3 March 2026
Location	Process building 2, pump bay B, level 0	**Document owner**	Reliability Engineer, Maintenance & Reliability
Applies to	Planned and corrective seal replacement	**Next review**	March 2027

SAFETY – READ FIRST. No work on P-207 may start without a valid **permit to work** and a signed **isolation certificate**. This manual does **not** contain the isolation procedure. Isolation points, sequence and verification are defined on the isolation certificate issued for the job (template IC-P207). If the permit or isolation certificate is missing or unclear, stop and contact the area authority.

7.1 Equipment data

Item	Value
Service	Hot process water circulation, loop 2
Pump type	End-suction, back-pull-out centrifugal pump, overhung impeller
Driver	Electric motor 37 kW, 2960 rpm, flexible spacer coupling
Medium	Treated process water, 75–85 °C, light suspended solids
Discharge pressure (normal / max)	4.2 bar(g) / 6.5 bar(g)
Shaft diameter at seal	45 mm
Mechanical seal	Single cartridge seal, balanced, size 45 mm
Seal faces	Silicon carbide vs. carbon (rotating SiC / stationary carbon)
Secondary seals (O-rings)	EPDM, peroxide-cured, suitable for hot water
Flush arrangement	API Plan 11 – recirculation from pump discharge via orifice (Ø 3 mm) to seal chamber
Seal spare part no.	VPP-SP-45-CS11 (complete cartridge). Seal repair kit: VPP-SP-45-RK
Planned seal life	36 months / ~25,000 operating hours

7.2 Tools, spares and consumables

- Replacement cartridge seal **VPP-SP-45-CS11** – check the part number and the seal size on the box label before work starts. Do not fit a seal from an unlabelled or damaged box.

- Dial indicator with magnetic base (resolution 0.01 mm) for runout and end-play checks.
- Calibrated torque wrench (10–150 Nm).
- Lint-free cloths, isopropyl alcohol, clean water.
- O-ring lubricant approved for EPDM (silicone-based). **Never use mineral oil or grease on EPDM O-rings.**
- New casing gasket and new coupling spacer bolts if the old ones show damage.
- Laser alignment tool for final coupling alignment.

7.3 Removal of the old seal (summary)

1. Confirm the permit to work and the isolation certificate are signed, and that the pump is drained and depressurised as stated on the certificate.
2. Remove the coupling guard and the coupling spacer. Mark the coupling halves for reference.
3. Disconnect the Plan 11 flush line at the seal gland.
4. Remove the bearing frame (back-pull-out) and move it to the workshop bench.
5. **Fit the setting clips on the old cartridge seal before loosening the set screws.** Loosen the drive collar set screws, remove the gland nuts and slide the cartridge off the shaft.
6. Keep the old seal for failure analysis. Do not clean the seal faces – label the seal with tag, date and work order number and hand it to the reliability engineer.

7.4 Checks before fitting a replacement seal

Most early seal failures on P-207 are caused by conditions that existed **before** the new seal was fitted. All checks below must be completed and recorded on the work order. If a value is outside tolerance, **do not fit the seal** – correct the cause first or contact the reliability engineer.

A. Shaft and bearings

#	Check	Acceptance criterion
A1	Shaft runout at the seal area (dial indicator, rotate shaft by hand)	≤ 0.05 mm TIR
A2	Axial end-play of the shaft (push/pull)	≤ 0.10 mm
A3	Radial movement of the shaft at the seal area (lift test)	≤ 0.05 mm
A4	Bearings: rotate by hand and listen for roughness or noise	Smooth, no noise. Replace bearings if in doubt
A5	Shaft surface under the O-ring: scratches, fretting, corrosion, old O-ring residue	Clean and smooth, Ra ≤ 0.8 µm, no grooves
A6	Shaft shoulders, keyways and threads the O-ring passes over	Deburred and chamfered, or covered with tape during fitting

B. Seal chamber and pump casing

#	Check	Acceptance criterion
B1	Seal chamber face perpendicular to the shaft (indicator on shaft, sweep the face)	≤ 0.03 mm
B2	Seal chamber bore concentric to the shaft	≤ 0.13 mm TIR
B3	Seal chamber and gland face: deposits, scale, damage to the gasket face	Clean, flat, undamaged
B4	Plan 11 flush line and orifice (Ø 3 mm)	Line clear, orifice not blocked or enlarged
B5	Impeller and wear rings: wear, rubbing marks	Wear ring clearance ≤ 0.6 mm (diametral)

C. The replacement seal

#	Check	Acceptance criterion
C1	Part number and size on the label match section 7.1	VPP-SP-45-CS11, 45 mm
C2	Seal faces: visible chips, cracks or scratches (do not touch the faces with bare fingers)	No damage. Faces clean and dry
C3	O-rings: correct material (EPDM), no cuts or flat spots	EPDM, undamaged
C4	Setting clips fitted and set screws not tightened	Clips in place as delivered
C5	Storage time and condition of the spare	Max. 5 years from production date, stored dry and dust-free

> **Note.** The most common causes of early failure recorded on P-207 are **dry running** after maintenance (pump not vented before start), **coupling misalignment** after the bearing frame is refitted, and a **blocked Plan 11 orifice**. Checks B4, 7.6 step 3 and 7.7 steps 1–3 address these causes directly.

7.5 Installation

1. Clean the shaft and seal chamber with isopropyl alcohol and lint-free cloth.
2. Apply a thin film of approved silicone-based lubricant to the shaft and the inner O-ring. **Do not lubricate the seal faces.**
3. Slide the cartridge onto the shaft with an even push. Do not use a hammer and do not twist the cartridge over keyways.
4. Refit the bearing frame to the casing with a new casing gasket. Tighten the casing bolts crosswise to the torque in section 7.8.
5. Fit the gland nuts and tighten them evenly in a crosswise sequence to the torque in section 7.8. The gland must sit square against the seal chamber face.
6. Tighten the drive collar set screws to the torque in section 7.8.
7. **Remove the setting clips** and store them on the pump for the next seal change. The shaft must turn freely by hand.
8. Reconnect the Plan 11 flush line at the gland connection marked **F** (flush). Do not connect to the connection marked **D** (drain/plug).

7.6 Coupling and final checks

1. Refit the coupling spacer.
2. Check motor and pump hold-down bolts for soft foot (≤ 0.05 mm).
3. Align the coupling with the laser alignment tool. Tolerance: **offset ≤ 0.05 mm, angularity ≤ 0.05 mm per 100 mm**. Record the final values on the work order.
4. Refit the coupling guard.

7.7 Commissioning

> **Do not start the pump dry.** A mechanical seal can be destroyed within seconds of dry running. Commissioning must only start when the area authority has handed back the equipment under the permit to work.

1. Fill the pump and the suction line with liquid.
2. Vent the pump casing and the seal chamber through the vent on the gland until liquid without air comes out.
3. Check that the Plan 11 flush line is open.
4. Check direction of rotation by a short bump start (only if the coupling has been disconnected from the motor).
5. Start the pump against a partly open discharge valve and open the valve slowly to the normal operating point.
6. Observe the seal for 30 minutes. A few drops during the first minutes is normal while the faces bed in. Visible leakage after 30 minutes is not acceptable.
7. Check gland temperature after 30 minutes: max. 25 °C above medium temperature.
8. Record vibration on the bearing housings: ≤ 2.8 mm/s RMS.

7.8 Tightening torques

Fastener	Size	Torque	Note
Gland nuts (seal cartridge to seal chamber)	M12	40 Nm	Crosswise, in two passes
Drive collar set screws	M8	10 Nm	Tighten before removing setting clips
Casing bolts (bearing frame to casing)	M16	120 Nm	Crosswise, new gasket
Coupling spacer bolts	M10	45 Nm	Replace if damaged
Pump and motor hold-down bolts	M20	230 Nm	Check soft foot first

7.9 Documentation

- Record all values from 7.4, 7.6 and 7.7 on the work order in the maintenance system.
- Attach photos of the old seal faces and the shaft surface.
- Hand over the old seal to the reliability engineer for failure analysis.
- Past seal failures and work orders for P-207 are found in the maintenance history on the Planning and Operations site, not in this manual.

Fictitious document created for training purposes. Vestbjerg Processing Plant, equipment tags, part numbers and values are examples only and must not be used for real maintenance work.